

Topic 5 - Arms Race

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Introduction

This section should outline the real-world background and context of your model. Discuss the motivation one might have to study this model. You may want to research the relevant literature and include some citations to support your framing of the real-world context (depending on the project). Be sure to include only background information and context that is directly related to your research project. Really focus on what the reader needs to know to understand your work, and be interested and excited by it.

The final paragraph of your Introduction should frame your research question and describe what you will be studying in your paper. Briefly mention, in words, what you will be doing and showing in the rest of the paper. It is often good to lay out your main finding/result here, too. A good Introduction section for an applied mathematics paper ideally has no mathematical symbols or jargon; instead, it should be focused on the real-world context and what you are trying to study. The math is a tool you will introduce later, not the main focus of the paper.

Base Model

For either hypothetical country, P and G, the assumptions for our base model are as follows:

1. The rate of change in one nation's arms spending is directly proportional to the other nation's arms spending
2. The rate of change in one nation's arms spending is also negatively directly proportional to that nation's own current spending
3. There may be underlying trust and hostility between the two nations, which will have a constant effect on rate of change
4. For now, we assume that each country has infinite amounts of wealth at their disposal (not realistic at all in a real-world context)

Based on these assumptions, the following model was derived:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r \quad (1)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s \quad (2)$$

In particular, $p(t), g(t)$ denote the arms spending at time t for countries P and G respectively. Meanwhile, $\frac{\partial p}{\partial t}, \frac{\partial g}{\partial t}$ denote the rate of change of each country's arm spending. The variables a, c measure the effect of mutual fear that each country has on the other's arms spending, b, d measure the effect of expenditure burden that each country has on their own spending, and r, s measure the underlying hostility or trust that the each nation has to another.

For the sake of realism, $p(t), g(t)$ are held to be non-negative values as negative arms spending would not make sense in this scenario.

Moreover, to ensure that mutual fear and expenditure burden match the behavior as laid out in the assumptions, a, b, c, d are held to be non-negative as well. On the other hand, r, s can assume any value where negative values denote trust and positive values denote hostility.

The parameters in the model (a, b, c, d, r, s) assume no form of units as they serve as abstractions of qualitative measures (i.e. it would be hard to attach a unit to expenditure burden or fear). However, the state variables $p(t), g(t)$ themselves would be expressed in some common currency (e.g. U.S. dollars) and the derivatives of those state variables $(\frac{\partial p}{\partial t}, \frac{\partial g}{\partial t})$ would be expressed in that currency over time (e.g. U.S. dollars per year).

Theoretically speaking, this model is a system of multivariate, autonomous linear differential equations. In this situation, the Jacobian would have to be computed for the purposes of stability analysis.

Analysis of Base Model

To facilitate the stability analysis of the base model, we now reformulate the system of equations into matrix form:

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix} + \begin{bmatrix} r \\ s \end{bmatrix}$$

Analysis without underlying trust or hostility

As a starting point, we assume that both nations have no underlying trust or hostility such that $r = s = 0$. Thus, the model equations are now:

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix}$$

Note that by having the system in linear form, the Jacobian is simply the matrix

$$J = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix}$$

Setting $r = s = 0$ lends a unique advantage in that the resulting system of linear differential equations is now homogenous (i.e. there is no constant term). This means that a closed-form of the state-variables can be found in the form of:

$$\begin{bmatrix} p(t) \\ g(t) \end{bmatrix} = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix} P^{-1} \begin{bmatrix} p(0) \\ g(0) \end{bmatrix}$$

Where λ_1, λ_2 denotes the eigenvalues of the Jacobian J , P denotes the matrix of eigenvectors, and $p(0), g(0)$ denotes the initial spending of the two countries.

To determine equilibrium points, set

$$\begin{bmatrix} \frac{\partial p}{\partial t} \\ \frac{\partial g}{\partial t} \end{bmatrix} = 0$$

Thus,

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix} \begin{bmatrix} p(t) \\ g(t) \end{bmatrix}$$

An obvious equilibrium that can be derived from this formulation is:

$$\begin{bmatrix} p^*(t) \\ g^*(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This equilibrium point corresponds to a scenario where both nations mutually demilitarize.

To determine the stability conditions of such a scenario, we need to compute the Jacobian. Since the system of differential equations are already in linear form, the Jacobian is simply given by:

$$J = \begin{bmatrix} -b & a \\ c & -d \end{bmatrix}$$

In order for the scenario of mutual disarmament to be asymptotically stable, the eigenvalues of the jacobian should both be negative. The eigenvalues are given by:

$$\det(J - I\lambda) = \begin{vmatrix} -b - \lambda & a \\ c & -d - \lambda \end{vmatrix} = \lambda^2 + (b + d)\lambda + bd - ac = 0$$

The solutions to these eigenvalues are of the form:

$$\lambda = \frac{-(b + d) \pm \sqrt{(b + d)^2 - 4(bd - ac)}}{2}$$

Expanding the square term in the discriminant and simplifying the equation yields the following:

$$\lambda = \frac{-(b+d) \pm \sqrt{(b-d)^2 + 4ac}}{2}$$

It should be noted that because of the non-negative restrictions imposed upon a, b, c, d , the discriminant will always remain positive. From this, we expect that the model will always behave monotonically. In other words, the arms spending of countries P and G will not oscillate.

To check when the eigenvalues remain negative, we look for conditions when $(b+d)$ is greater than $\sqrt{(b-d)^2 + 4ac}$ in magnitude. After doing some algebra, the result turns out to be: $bd > ac$. In the context of the scenario, the arms spending of countries P and G will converge to complete mutual disarmament if their combined expenditure burden outweighs the combined mutual fear they have for each other.

On the other hand, if $bd < ac$, then the combined mutual fear will outweigh the combined expenditure burden. In this case, the equilibrium of mutual disarmament will become unstable and both nations will engage in a perpetual arms race.

Another point to consider is what happens when $bd = ac$. In this case, we substitute $ac = bd$ in the formula for the eigenvalues:

$$\lambda = \frac{-(b+d) \pm \sqrt{(b+d)^2 - 4(bd - ac)}}{2} \lambda = 0, -(b+d)$$

In this case, one eigenvalue is negative and the other is 0. The closed form sheds further light onto what this implies for the model. To this end, the eigenvalues were substituted back in the Jacobian to derive the eigenvectors.

These eigenvectors are

$$\tilde{v}_1 = \begin{bmatrix} \frac{a}{b} \\ 1 \end{bmatrix}$$

for $\lambda_1 = 0$ and

$$\tilde{v}_2 = \begin{bmatrix} -\frac{a}{d} \\ 1 \end{bmatrix}$$

for $\lambda_2 = -(b + d)$.

The closed form solution is therefore:

$$\begin{bmatrix} p(t) \\ g(t) \end{bmatrix} = \begin{bmatrix} \frac{a}{b} & -\frac{a}{d} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e^{(0)t} & 0 \\ 0 & e^{-(b+d)t} \end{bmatrix} \begin{bmatrix} \frac{a}{b} & -\frac{a}{d} \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} p(0) \\ g(0) \end{bmatrix}$$

To determine the long-term behavior of the closed form, we let $t \rightarrow \infty$, which results in the following convergence.

$$\begin{bmatrix} p(\infty) \\ g(\infty) \end{bmatrix} = \begin{bmatrix} \frac{dp(0)+ag(0)}{b+d} \\ \frac{cp(0)+bg(0)}{b+d} \end{bmatrix}$$

Within the scope of the scenario, this convergence suggests that both countries will reach a stabilized level of spending contingent on what they were initially spending.

Summary

To summarize the analysis, the base model without any underlying trust or hostility has three types of convergences. Attached alongside each explanation of these convergences are vector plots to illustrate the scenarios. These vector plots were constructed using R. For each R simulation, the parameter values are listed in the title of each plot. The vector plot shows the

direction of the nations' arms spending for each possible point in the simulation (showcased by the arrows). The red lines indicate the possible trajectories that each nation's arm spending can take based on their initial spending.

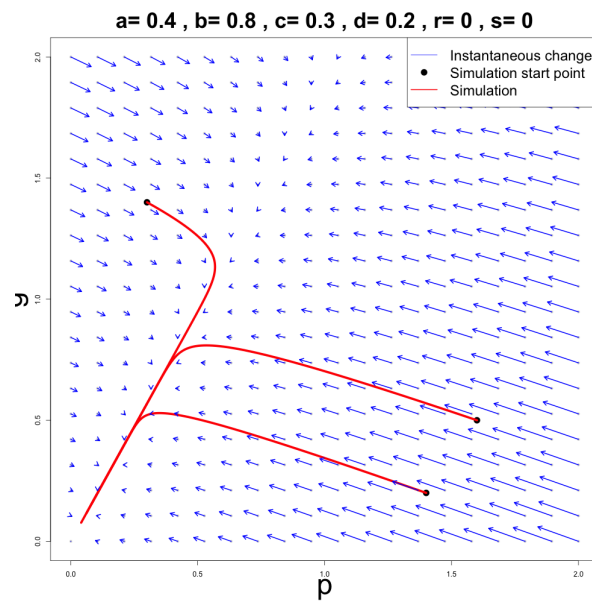


Figure 1: Mutual Disarmament

When $bd > ac$, both countries mutually disarm for various initial spending points ((0,0) is a stable equilibrium point). Figure 1 indicates that all the paths in this scenario reach (0,0)

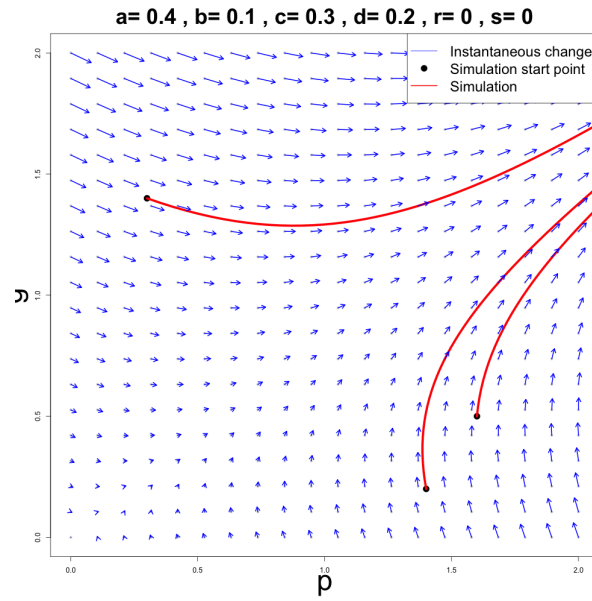


Figure 2: Perpetual Arms Race

When $bd < ac$, both countries engage in an a perpetual arms race ((0,0) is an unstable equilibrium point). Figure 2 shows each simulation path veering away from (0,0) and going beyond the borders of the graph.

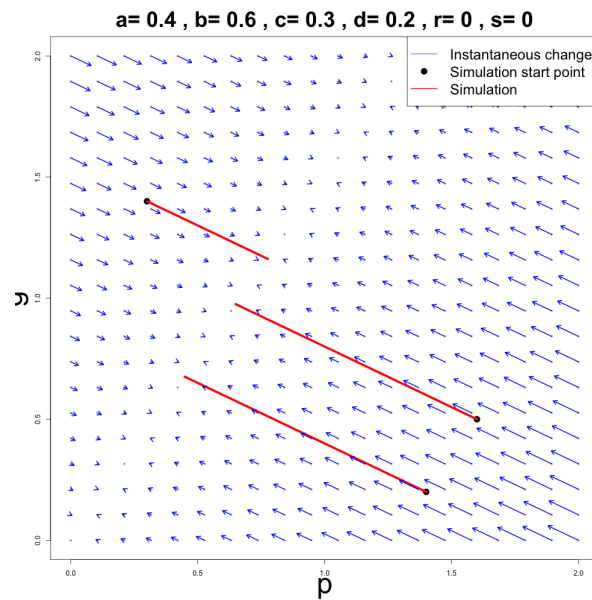


Figure 3: Stable Spending

When $bd = ac$, both countries reach a stabilized level of spending contingent on initial

spending (There is a line of stable equilibrium points). Figure 3 shows each path reaching to a stable line in a straight trajectory.

Analysis with trust or hostility

Now that we assume r, s to be some other value other than zero, the model is not homogenous anymore, meaning that there is not a convenient closed form solution of the system as there was before.

The equilibrium point also changes:

$$\begin{bmatrix} p^*(t) \\ g^*(t) \end{bmatrix} = \begin{bmatrix} \frac{rd+sa}{bd-ac} \\ \frac{rc+sb}{bd-ac} \end{bmatrix}$$

Now the convergences of the model may not necessarily converge to zero. To determine the stability conditions of this equilibrium, the Jacobian from the previous form of the model can be reused as the system is still linear. In this case, we have the same stability conditions for the equilibrium point as before: $bd > ac$.

To determine what happens when $bd = ac$, we conduct simulations to adjust the values of r and s . We uncover four cases when we loosen r and s conditions, allowing our trust/hostility parameters to take on $r, s \neq 0$ values. For each case, we provide an explanation accompanied by a vector plot which visualizes convergence. Vector plots were constructed using R, we provide the parameter values in the title of each plot for each case. The vector plot shows the direction of the nations' arms spending for each possible point in the simulation (showcased by the arrows). The red lines indicate the possible trajectories that each nation's arm spending can take based on their initial spending.

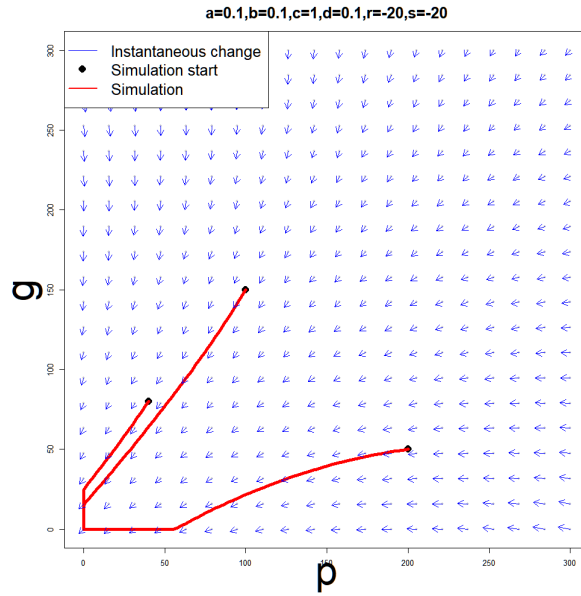


Figure 4: Mutual Disarmament

We observe mutual disarmament in Figure 4, in which spending for all trajectories decays to $(0,0)$ for both nations, when $r < 0$ and $s < 0$. This represents opposing nations trusting each other. This first case is most similar to the mutual disarmament seen when r and s are zero; however, this mutual disarmament is solely driven by trust between Nations P and G.

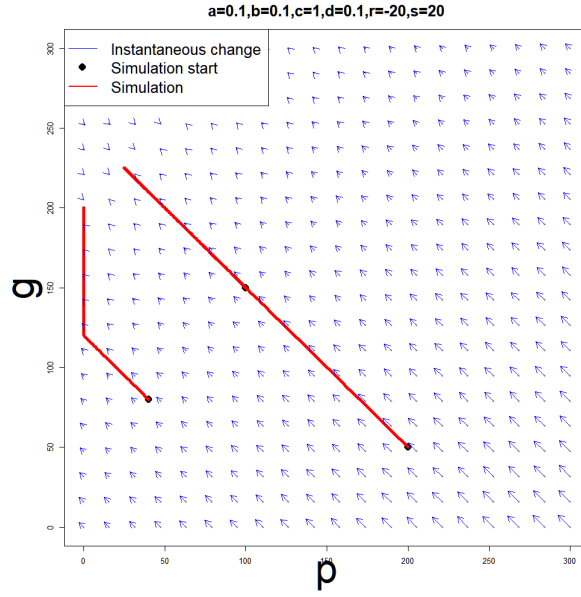


Figure 5: Disarmament and Aramament

When r and s are of opposing sign, we find that one nation is conciliatory and will disarm, decaying to $(0,0)$, while the other nation is hostile and will arm itself (with bound, not infinitely) for all trajectories. In Figure 5, we set $r < 0$ and $s > 0$ and observe that, for all trajectories, Nation P fully disarms, while Nation G converges to a bounded value depending on the initial conditions of the system and system parameters.

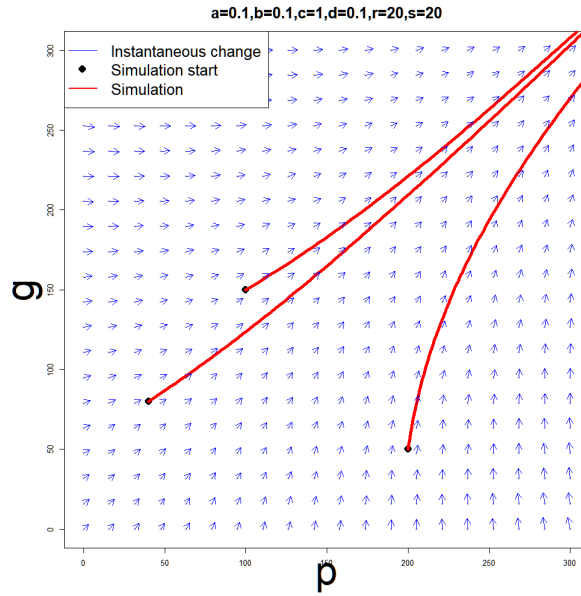


Figure 6: Symmetric Perpetual Arms Race

We observe a perpetual arms race in Figure 6, in which spending grows without bound for both nations, when $r > 0$ and $s > 0$. This represents that opposing nations hold independent grievances. Arms spending will diverge the quickest as Nations P and G arm independently and reactively to the other nation arming itself. It is important to note that this perpetual armament is balanced; as one can observe, the red trajectory lines converge in a symmetric fashion relative to x- and y-axes.

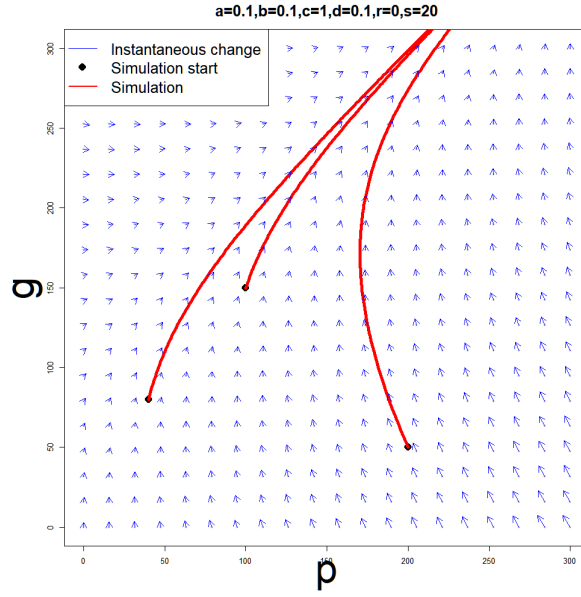


Figure 7: Asymmetric Perpetual Arms Race

We observe a perpetual arms race in Figure 7, in which spending grows without bound for both nations. When one of r and s is set to zero (no feelings towards the opposing nation), and the other is set to a positive value (hostile feelings towards the opposing nation). Unlike in the previous perpetual arms race, this arms race is unbalanced; one nation drives the arms race while the other reacts and arms accordingly. In this case, we set $r = 0$ and $s > 0$ and observe that Nation G drives the arms race while Nation P lags behind in spending; the red trajectory converge in an asymmetric fashion relative to x- and y-axes, growing more quickly in the y-axis (Nation G arms spending).

We believe that the inclusion of $r, s \neq 0$ possibilities, we more accurately predict how real-world nations react to the (dis)armament of their contemporaries. With $r, s \neq 0$, we can account for goodwill and grievance between nations, where in previous cases with parameters r and s excluded, we could only predict independent behaviour. This model still has flaws, however. In particular, nations are able to arm themselves without bound, which is not realistic as money is a finite resource. We hope that our model extension will correct the issues that

our base model is restricted by.

Model Extension

Mutual Logistic Fear Model

One major limitation of the base model is the assumption that each country has access to infinite resources. In reality, arms spending is constrained by each country's limited amounts of funds. As a result, the base model produces unrealistic outcomes such as unbounded growth in arms expenditure. To address this limitation, we extend the model by introducing logistic growth terms.

Model overview

For either hypothetical country P and G, the assumptions for the extended model are as follows:

1. The rate of change in one nation's arms spending follows a logistic growth depending on the other nation's arms spending. This is accompanied by a finite resource constraint that limits how much each country can spend.
2. The rate of change is negatively proportional to the nation's own current spending.
3. There may still be underlying trust or hostility between the two nations, which has a constant effect on the rate of change.

Based on these assumptions, the following model was formulated:

$$\frac{\partial p}{\partial t} = a \left(1 - \frac{p(t)}{K_p} \right) g(t) - bp(t) + r \quad (3)$$

$$\frac{\partial g}{\partial t} = c \left(1 - \frac{g(t)}{K_g} \right) p(t) - dg(t) + s \quad (4)$$

Here, K_p, K_g denote the maximum level of sustainable arms spending for countries P and G respectively. The units these parameters assume would be in a common currency such as U.S. dollars. The rest of the parameters and state variables assume the same meaning as specified for the base model.

Differences from the Base Model

The extended model differs from the base model in one fundamental way: the inclusion of the logistic terms $\left(1 - \frac{p}{K_p} \right)$ and $\left(1 - \frac{g}{K_g} \right)$.

In particular:

- The base model allows for unbounded growth when mutual fear dominates.
- The extended model prevents unbounded growth by enforcing resource constraints.

This extension therefore provides a more realistic representation of arms dynamics by combining strategic interaction with physical and economic limitations.

Analysis of Mutual Logistic Fear Model

Jacobian Matrix

To determine the local stability of an equilibrium point, we compute the Jacobian matrix:

$$J(p, g) = \begin{bmatrix} \frac{\partial f}{\partial p} & \frac{\partial f}{\partial g} \\ \frac{\partial h}{\partial p} & \frac{\partial h}{\partial g} \end{bmatrix}$$

After computing the partial derivatives, the Jacobian of the extended model is

$$J(p, g) = \begin{bmatrix} -\frac{ag}{K_p} - b & a \left(1 - \frac{p}{K_p}\right) \\ c \left(1 - \frac{g}{K_g}\right) & -\frac{cp}{K_g} - d \end{bmatrix}$$

Evaluating this Jacobian at an equilibrium point (p^*, g^*) yields

$$J(p^*, g^*) = \begin{bmatrix} -\frac{ag^*}{K_p} - b & a \left(1 - \frac{p^*}{K_p}\right) \\ c \left(1 - \frac{g^*}{K_g}\right) & -\frac{cp^*}{K_g} - d \end{bmatrix}$$

The eigenvalues of this matrix determine the local stability of the equilibrium.

Equilibrium Points

We set:

$$a \left(1 - \frac{p^*}{K_p}\right) g^* - bp^* + r = 0$$

$$c \left(1 - \frac{g^*}{K_g}\right) p^* - dg^* + s = 0.$$

These equations define the equilibrium point (p^*, g^*) .

Rearranging the first equation yields:

$$a \left(1 - \frac{p^*}{K_p}\right) g^* = bp^* - r,$$

and hence,

$$g^* = \frac{K_p(bp^* - r)}{a(K_p - p^*)}, \quad p^* \neq K_p.$$

Similarly, rearranging the second equation gives:

$$c \left(1 - \frac{g^*}{K_g} \right) p^* = dg^* - s,$$

and thus,

$$p^* = \frac{K_g(dg^* - s)}{c(K_g - g^*)}, \quad g^* \neq K_g.$$

Stability Conditions

For a two-dimensional system, an equilibrium point is locally asymptotically stable if the Jacobian evaluated at that point has:

$$\text{tr}(J(p^*, g^*)) < 0 \quad \text{and} \quad \det(J(p^*, g^*)) > 0.$$

Trace

The trace is

$$\text{tr}(J(p^*, g^*)) = \left(-\frac{ag^*}{K_p} - b \right) + \left(-\frac{cp^*}{K_g} - d \right).$$

So,

$$\text{tr}(J(p^*, g^*)) = -\frac{ag^*}{K_p} - \frac{cp^*}{K_g} - (b + d).$$

Since $a, b, c, d, K_p, K_g, p^*, g^* \geq 0$, it follows that $\text{tr}(J(p^*, g^*)) < 0$ for any nonnegative equilibrium point, provided $b + d > 0$.

Thus, in the extended model, the trace is automatically negative under the natural assumptions of the scenario.

Determinant

The determinant is

$$\det(J(p^*, g^*)) = \left(-\frac{ag^*}{K_p} - b\right) \left(-\frac{cp^*}{K_g} - d\right) - a \left(1 - \frac{p^*}{K_p}\right) c \left(1 - \frac{g^*}{K_g}\right).$$

Expanding this gives

$$\det(J(p^*, g^*)) = \left(b + \frac{ag^*}{K_p}\right) \left(d + \frac{cp^*}{K_g}\right) - ac \left(1 - \frac{p^*}{K_p}\right) \left(1 - \frac{g^*}{K_g}\right).$$

Unlike the base model, the determinant of the Jacobian in the extended model depends explicitly on the equilibrium values p^* and g^* .

As a result, it is generally not possible to derive a simple analytical condition for stability in terms of the parameters alone.

Eigenvalues and Stability

The eigenvalues of the Jacobian are given by

$$\lambda = \frac{\text{tr}(J(p^*, g^*)) \pm \sqrt{\text{tr}(J(p^*, g^*))^2 - 4 \det(J(p^*, g^*))}}{2}.$$

Clearly, this expression depends on the equilibrium values p^* and g^* , which are not generally available explicitly. Consequently, the eigenvalues do not yield simple analytical conditions for stability, unlike in the base model.

Simulations

We consider the following three representative parameter regimes while keeping $K_p = K_g = 5$ fixed.

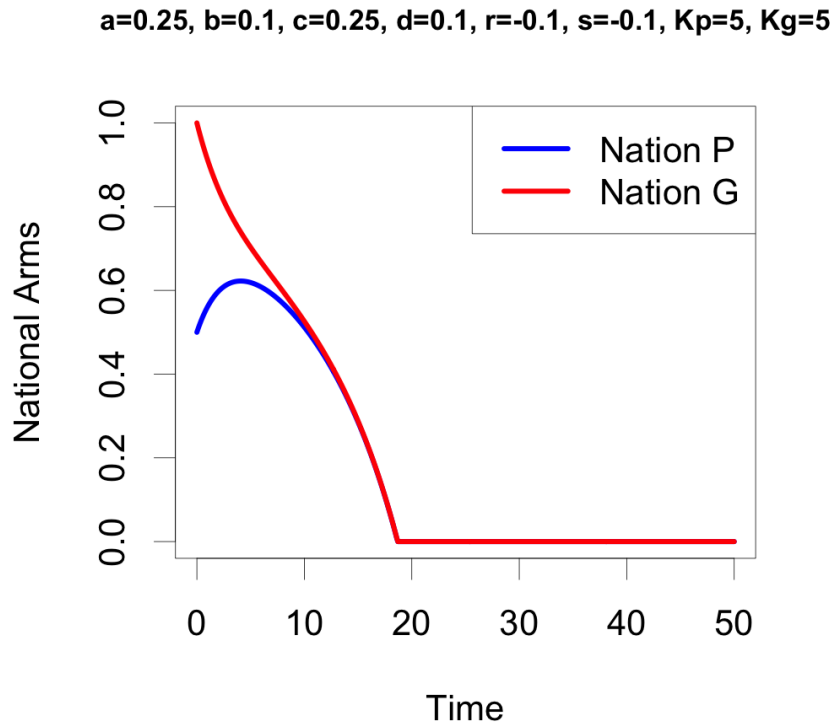


Figure 8: Case 1: Mutual disarmament

Case 1: $bd < ac, r < 0, s < 0$ In this case, both countries experience negative external pressure, representing trust or incentives to de-escalate. Arms spending falls over time, leading to mutual disarmament.

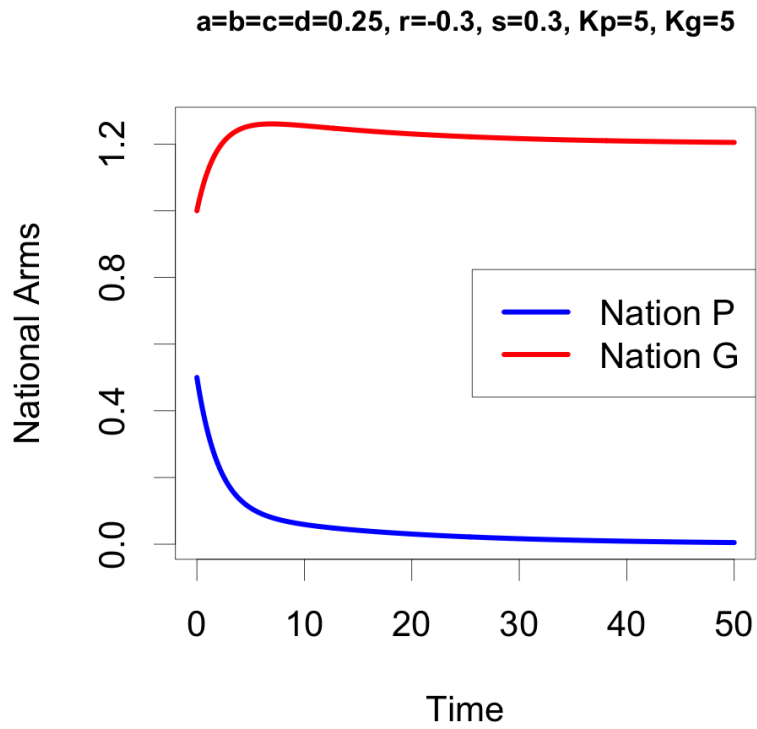


Figure 9: Case 2: Stabilized but unequal spending

Case 2: $bd = ac, r < 0, s > 0$ Here, the two countries face asymmetric external pressures. Country P experiences trust or downward pressure, while country G experiences hostility or upward pressure. The model predicts stabilization, but at unequal long-run spending levels.

$a=0.1, b=0.25, c=0.1, d=0.25, r=0.3, s=0.3, K_p=5, K_g=5$

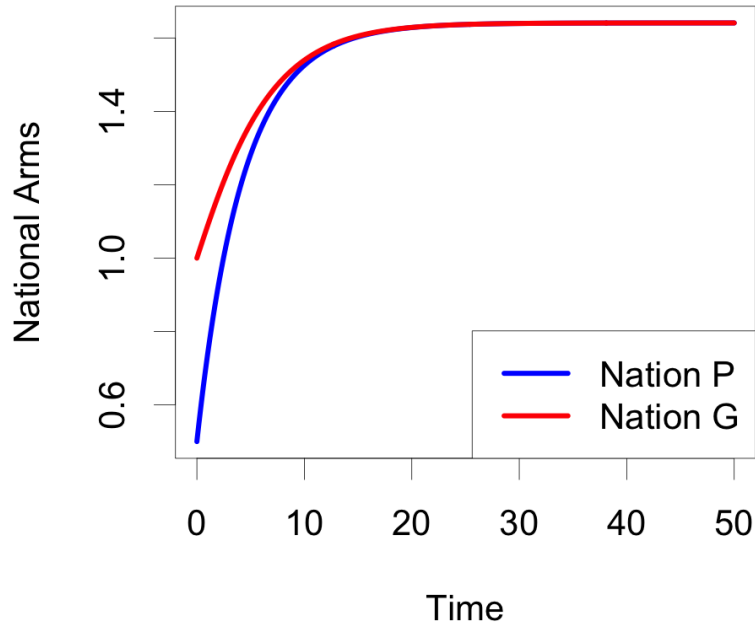


Figure 10: Case 3: High steady-state spending

Case 3: $bd > ac, r > 0, s > 0$ In this case, both countries experience positive external pressure, representing hostility or incentives to increase arms. Both countries converge to a high steady-state level of arms spending.

Results

Sensitivity Analysis

With regards to the sensitivity, we tested how changes in each of the variables in our model would impact nations arms spending. Here, only a, b , and r will be tested, but the conclusions relate to c, d and s in the same way respectively.

Sensitivity of a :

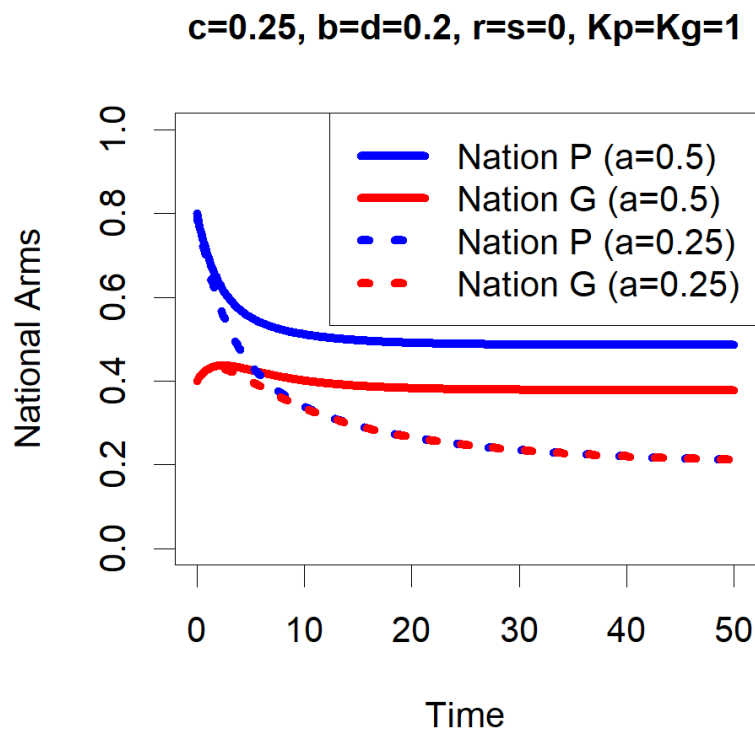


Figure 11: Sensitivity of a

In this case, we can see that increases in one country's **mutual fear (a)** will lead to increases in both countries' spending at all times. This is due to the fact that a higher a will lead to country **P** wanting to spend more for each unit of **G**'s arms, so **P**'s arms will be higher. Since $c > 0$ (model definition), this higher level of **P**'s arms will also lead to higher arms spending by **G**.

Sensitivity of b :

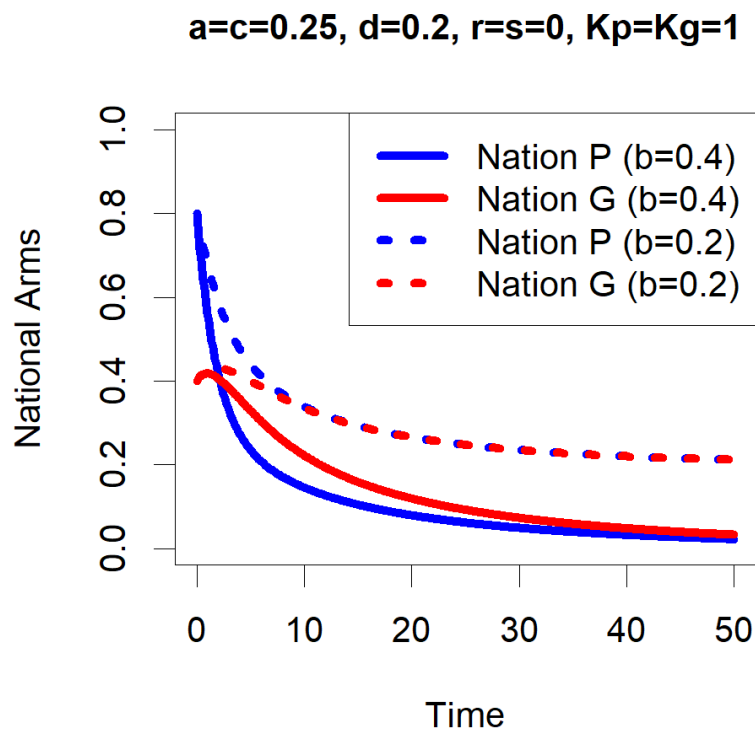


Figure 12: Sensitivity of b

In this case, we can see that increases in one country's **expenditure burden (b)** will lead to decreases in both countries spending at all times, this is due to the fact that a higher b will lead to $\mathbf{P}$ wanting to spend less for each unit of $\mathbf{P}$'s arms, so $\mathbf{P}$'s arms will be lower. Since $c > 0$ (model definition), this lower level of $\mathbf{P}$'s arms will also lead to lower $\mathbf{G}$'s arms spending

Sensitivity of r :

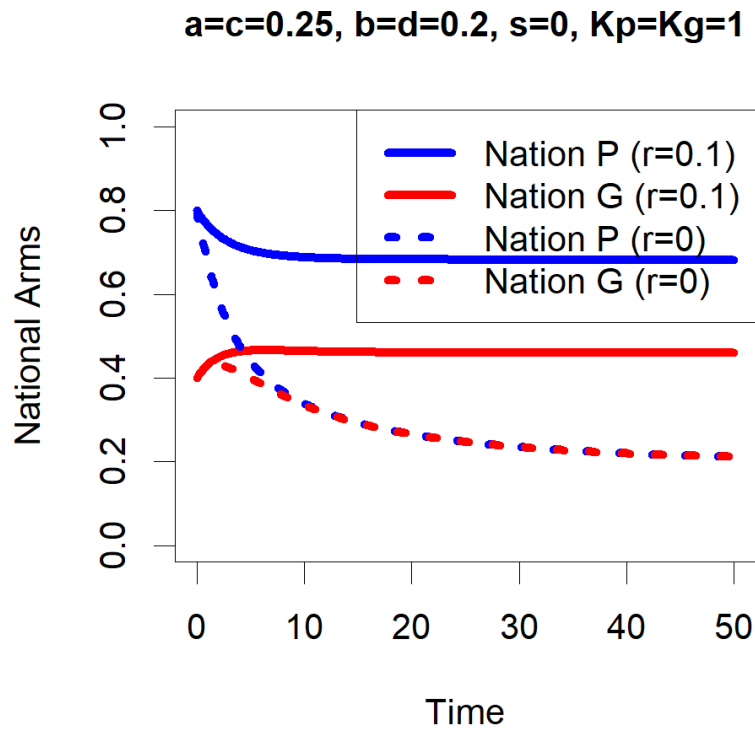


Figure 13: Sensitivity of r

In this case, we can see that increases in one country's **trust/hostility** (r) will lead to increases in both countries' spending at all times. This is due to the fact that a higher r will lead to **P** wanting to spend more due to its higher grievance towards **G**, so **P**'s arms will be higher. Since $c > 0$ (model definition), this higher level of **P** arms will also lead to higher **G** arms spending.

Discussion

This is the final section of the main body of your paper. Start by briefly summarizing the main results you presented in the last section and relating them back to your

research question. This is also where you should be providing answers, based on your results, to your research question(s). Be sure to interpret your results in the context of the real-world scenario framing your project. What are the implications of your results?

You should also discuss limitations of your work. This is usually related to the modelling assumptions you made before. When are your results applicable and when are they not applicable? What factors have you omitted from your model that may affect the predictions? This is important to discuss, otherwise the reader may come away from your paper with the impression that your results are universal which is rarely the case.

Finally, you should discuss possible directions for future research. Did your results open up any other questions? Where else could you go with this work? Are there other things you've read about in the literature that could be interesting to look at? You should also use this as a way to conclude your whole paper.

Appendix: Individual Contributions

Zeyn Jaswal

1. Presentation: Overview
2. Report: Introduction and Conclusion

Azzaam Khan

1. Presentation: Base Model
2. Report: Base Model

Timothy Palin

1. Presentation: Results
2. Report: Results

Tony Xu

1. Presentation: Logistic Mutual Fear Extension
2. Report: Logistic Mutual Fear Extension

Meredith Reeves

1. Presentation: Public Sentiment
2. Report: Public Sentiment

References